

Esp32Tap Breadboard — Cluster-Sorted Point-to-Point Wiring

Follow the Build Step column from top to bottom. Ref preserves the original 1–126 numbering. Map the RJ45 breakout terminals and SOT-23 adapters with a meter before using their labels.

Start with all power disconnected. Leave STANDALONE POWER and COIL POWER removed. The relay datasheet pin drawing is a bottom view—meter-verify K1 before wiring.

Wire colors: BLACK RED ORANGE BLUE VIOLET YELLOW GREEN WHITE NO WIRE

Build 1 — Cluster 3: ESP32 DevKit						
Step	Ref	Color	From	To	Part	Note
1	29	BLUE	DevKit 3V3	LOGIC_3V3	Jumper wire	Logic rail
2	30	BLACK	DevKit GND	GND	Jumper wire	Common ground

Build 2 — Cluster 7: Feedback, VBUS, and LEDs						
Step	Ref	Color	From	To	Part	Note
3	105	BLACK	K1 pin 11	GND	Wire	Feedback COM
4	106	GREEN	K1 pin 13	DevKit GPIO4	Wire	NC feedback
5	107	BLUE	LOGIC_3V3	R_FB_NC 10 kΩ, end 1	Wire	NC pull-up
6	108	GREEN	R_FB_NC end 2	DevKit GPIO4	Wire	NC pull-up node
7	109	GREEN	K1 pin 9	DevKit GPIO5	Wire	NO feedback
8	110	BLUE	LOGIC_3V3	R_FB_NO 10 kΩ, end 1	Wire	NO pull-up
9	111	GREEN	R_FB_NO end 2	DevKit GPIO5	Wire	NO pull-up node
10	112	VIOLET	DevKit 5V/VBUS	2N7000 gate	Wire	Verify physical lead
11	113	VIOLET	DevKit 5V/VBUS	R_VBUS_DISCHARGE 10 kΩ, end 1	Wire	Gate discharge
12	114	BLACK	R_VBUS_DISCHARGE end 2	GND	Wire	Discharge return
13	115	BLACK	2N7000 source	GND	Wire	Verify physical lead
14	116	GREEN	2N7000 drain	DevKit GPIO7	Wire	VBUS_PRESENT_N
15	117	BLUE	LOGIC_3V3	R_VBUS_PULLUP 10 kΩ, end 1	Wire	Drain pull-up
16	118	GREEN	R_VBUS_PULLUP end 2	DevKit GPIO7	Wire	Pull-up node
17	119	YELLOW	DevKit GPIO38	R_STATUS 330 Ω, end 1	Wire	Status LED
18	120	YELLOW	R_STATUS end 2	Green LED anode	Wire	Long LED lead
19	121	BLACK	Green LED cathode	GND	Wire	Short/flat-side lead
20	122	BLUE	LOGIC_3V3	R_POWER_LED 1 kΩ, end 1	Wire	Power LED
21	123	YELLOW	R_POWER_LED end 2	Red LED anode	Wire	Long LED lead
22	124	BLACK	Red LED cathode	GND	Wire	Short/flat-side lead
23	125	BLUE	LOGIC_3V3	C_LOGIC 100 nF, end 1	Wire	Logic rail bypass
24	126	BLACK	C_LOGIC end 2	GND	Wire	Bypass return

Build 3 — Cluster 6: Hardware gating and UART

Step	Ref	Color	From	To	Part	Note
25	58	BLUE	LOGIC_3V3	AHC08 pin 14	Wire	VCC
26	59	BLACK	AHC08 pin 7	GND	Wire	Ground
27	60	YELLOW	DevKit GPIO21	AHC08 pin 1	Wire	RELAY_CMD
28	61	YELLOW	DevKit GPIO21	R_RELAY_CMD_PD 10 kΩ, end 1	Wire	Command pull-down
29	62	BLACK	R_RELAY_CMD_PD end 2	GND	Wire	Pull-down return
30	63	YELLOW	TREAD_OK	AHC08 pin 2	Wire	Relay permission
31	64	YELLOW	AHC08 pin 3	RELAY_GATE	Wire	Relay gate output
32	65	YELLOW	DevKit GPIO15	AHC08 pin 4	Wire	TX_ENABLE
33	66	YELLOW	DevKit GPIO15	R_TX_ENABLE_PD 10 kΩ, end 1	Wire	Enable pull-down
34	67	BLACK	R_TX_ENABLE_PD end 2	GND	Wire	Pull-down return
35	68	YELLOW	TREAD_OK	AHC08 pin 5	Wire	TX permission
36	69	YELLOW	AHC08 pin 6	TX_GATE	Wire	TX gate output
37	70	BLACK	AHC08 pins 9,10,12,13	GND	Four wires	Tie unused inputs low
38	71	NO WIRE	AHC08 pins 8,11	NO CONNECTION	—	Unused outputs
39	72	BLUE	AHC08 pin 14	C_AHC08 100 nF, end 1	Wire	Place close to IC
40	73	BLACK	C_AHC08 end 2	AHC08 pin 7	Wire	Decoupling return
41	92	BLUE	LOGIC_3V3	AHC126 pin 14	Wire	VCC
42	93	BLACK	AHC126 pin 7	GND	Wire	Ground
43	94	YELLOW	TX_GATE	AHC126 pin 1	Wire	Gate 1 OE
44	95	YELLOW	DevKit GPIO17	AHC126 pin 2	Wire	Gate 1 input
45	96	YELLOW	AHC126 pin 3	R_TX 100 Ω, end 1	Wire	Gate 1 output
46	97	BLACK	AHC126 pins 4,5,9,10,12,13	GND	Six wires	Unused enables/inputs low
47	98	NO WIRE	AHC126 pins 6,8,11	NO CONNECTION	—	Unused outputs
48	99	BLUE	AHC126 pin 14	C_AHC126 100 nF, end 1	Wire	Place close to IC
49	100	BLACK	C_AHC126 end 2	AHC126 pin 7	Wire	Decoupling return
50	101	GREEN	CONSOLE RJ45 pin 6	R_CONS_RX 10 kΩ, end 1	Wire	Console RX tap
51	102	GREEN	R_CONS_RX end 2	DevKit GPIO18	Wire	Console RX
52	103	GREEN	RJ45 pin-3 net	R_PIN3_RX 10 kΩ, end 1	Wire	Pin-3 tap
53	104	GREEN	R_PIN3_RX end 2	DevKit GPIO16	Wire	Pin-3 RX

Build 4 — Cluster 1: RJ45 and relay contacts						
Step	Ref	Color	From	To	Part	Note
54	1	BLACK	CONSOLE RJ45 pin 1	GND	Jumper wire	First ground
55	2	BLACK	MOTOR RJ45 pin 1	GND	Jumper wire	First ground
56	3	BLACK	CONSOLE RJ45 pin 7	GND	Jumper wire	Second ground
57	4	BLACK	MOTOR RJ45 pin 7	GND	Jumper wire	Second ground
58	5	RED	CONSOLE RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact
59	6	RED	MOTOR RJ45 pin 2	+8V_RAW	Jumper wire	First +8 V contact
60	7	RED	CONSOLE RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact
61	8	RED	MOTOR RJ45 pin 8	+8V_RAW	Jumper wire	Second +8 V contact
62	9	WHITE	CONSOLE RJ45 pin 3	MOTOR RJ45 pin 3	Jumper wire	Direct pass-through
63	10	WHITE	CONSOLE RJ45 pin 4	MOTOR RJ45 pin 4	Jumper wire	Direct pass-through
64	11	WHITE	CONSOLE RJ45 pin 5	MOTOR RJ45 pin 5	Jumper wire	Direct safety pass-through
65	12	YELLOW	CONSOLE RJ45 pin 6	K1 G5V-2 pin 4	Jumper wire	Relay NC contact
66	13	YELLOW	MOTOR RJ45 pin 6	K1 G5V-2 pin 6	Jumper wire	Relay COM contact
67	14	YELLOW	K1 G5V-2 pin 8	R_TX 100 Ω, end 2	Jumper wire	Relay NO / ESP transmit

Build 5 — Cluster 2: Protected local power						
Step	Ref	Color	From	To	Part	Note
68	15	RED	+8V_RAW	RXEF075 lead 1	Wire	Local electronics tap only
69	16	RED	RXEF075 lead 2	1N5822 anode	Wire	Non-striped diode end
70	17	ORANGE	1N5822 cathode	VIN	Wire	Striped diode end
71	18	ORANGE	VIN	P6KE12A cathode	Wire	Striped TVS end
72	19	BLACK	P6KE12A anode	GND	Wire	TVS return
73	20	ORANGE	VIN	TSR1-2433E pin 1	Wire	Converter input
74	21	BLACK	TSR1-2433E pin 2	GND	Wire	Converter ground
75	22	BLUE	TSR1-2433E pin 3	TSR_3V3	Wire	Converter output
76	23	ORANGE	VIN	22 μF positive	Wire	Bulk capacitor
77	24	BLACK	22 μF negative	GND	Wire	Observe polarity
78	25	ORANGE	VIN	10 μF input-cap positive	Wire	Near TSR input
79	26	BLACK	10 μF input-cap negative	GND	Wire	Observe polarity
80	27	BLUE	TSR_3V3	10 μF output-cap positive	Wire	Near TSR output
81	28	BLACK	10 μF output-cap negative	GND	Wire	Observe polarity
82	31	BLUE	TSR_3V3	LOGIC_3V3	STANDALONE POWER jumper	Leave removed when USB is attached

Build 6 — Cluster 4: TPS3700 voltage monitor

Step	Ref	Color	From	To	Part	Note
83	32	ORANGE	VIN	TPS3700 pin 5 VDD	Wire	Supply
84	33	BLACK	TPS3700 pin 2	GND	Wire	Ground
85	34	ORANGE	VIN	R_UV_TOP 150 kΩ, end 1	Wire	UV divider
86	35	WHITE	R_UV_TOP end 2	UV_SENSE	Wire	Divider midpoint
87	36	WHITE	UV_SENSE	TPS3700 pin 3 INA+	Wire	UV input
88	37	WHITE	UV_SENSE	R_UV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom
89	38	BLACK	R_UV_BOTTOM end 2	GND	Wire	Divider return
90	39	WHITE	UV_SENSE	C_UV 1 nF, end 1	Wire	Filter
91	40	BLACK	C_UV end 2	GND	Wire	Filter return
92	41	ORANGE	VIN	R_OV_TOP 255 kΩ, end 1	Wire	OV divider
93	42	WHITE	R_OV_TOP end 2	OV_SENSE	Wire	Divider midpoint
94	43	WHITE	OV_SENSE	TPS3700 pin 4 INB-	Wire	OV input
95	44	WHITE	OV_SENSE	R_OV_BOTTOM 10 kΩ, end 1	Wire	Divider bottom
96	45	BLACK	R_OV_BOTTOM end 2	GND	Wire	Divider return
97	46	WHITE	OV_SENSE	C_OV 1 nF, end 1	Wire	Filter
98	47	BLACK	C_OV end 2	GND	Wire	Filter return
99	48	YELLOW	TPS3700 pin 1 OUTA	TREAD_OK	Wire	Open-drain output
100	49	YELLOW	TPS3700 pin 6 OUTB	TREAD_OK	Wire	Tie outputs together
101	50	BLUE	LOGIC_3V3	R_TREAD_PULLUP 10 kΩ, end 1	Wire	Pull-up
102	51	YELLOW	R_TREAD_PULLUP end 2	TREAD_OK	Wire	Pull-up node
103	52	YELLOW	TREAD_OK	R_TREAD_PULLDOWN 100 kΩ, end 1	Wire	Power-off pull-down
104	53	BLACK	R_TREAD_PULLDOWN end 2	GND	Wire	Pull-down return
105	54	YELLOW	TREAD_OK	R_TREAD_MCU 4.7 kΩ, end 1	Wire	GPIO isolation
106	55	YELLOW	R_TREAD_MCU end 2	DevKit GPIO6	Wire	TREAD_OK_MCU
107	56	ORANGE	VIN	C_TPS3700 100 nF, end 1	Wire	Decoupling
108	57	BLACK	C_TPS3700 end 2	GND	Wire	Decoupling return

Build 7 — Cluster 5: Relay supply and driver

Step	Ref	Color	From	To	Part	Note
109	74	ORANGE	VIN	TPS709 pin 1 IN	Wire	Regulator input
110	75	BLACK	TPS709 pin 2	GND	Wire	Ground
111	76	YELLOW	RELAY_GATE	TPS709 pin 3 EN	Wire	Hardware-gated enable
112	77	NO WIRE	TPS709 pin 4	NO CONNECTION	—	NC pin
113	78	VIOLET	TPS709 pin 5 OUT	+5V_RLY	Wire	Relay supply
114	79	ORANGE	VIN	C_TPS709_IN 1 μF, end 1	Wire	Input bypass
115	80	BLACK	C_TPS709_IN end 2	GND	Wire	Bypass return
116	81	VIOLET	+5V_RLY	C_TPS709_OUT 4.7 μF positive	Wire	Output capacitor
117	82	BLACK	C_TPS709_OUT negative	GND	Wire	Output return
118	83	YELLOW	RELAY_GATE	R_BASE 560 Ω, end 1	Wire	Base resistor
119	84	YELLOW	R_BASE end 2	BC337 base	Wire	Verify physical lead
120	85	YELLOW	BC337 base	R_BASE_PD 10 kΩ, end 1	Wire	Base pull-down
121	86	BLACK	R_BASE_PD end 2	GND	Wire	Pull-down return
122	87	BLACK	BC337 emitter	GND	Wire	Verify physical lead
123	88	VIOLET	BC337 collector	K1 pin 16	Wire	RELAY_COIL–
124	89	VIOLET	+5V_RLY	K1 pin 1	COIL POWER jumper	Leave removed initially
125	90	VIOLET	K1 pin 1	P6KE6.8CA lead 1	Wire	Bidirectional coil TVS
126	91	VIOLET	P6KE6.8CA lead 2	K1 pin 16	Wire	Bidirectional coil TVS

H. Before power

Check	Expected result
K1 pins 4 to 6	Continuity with COIL POWER removed
K1 pins 8 to 6	Open with COIL POWER removed
+8V_RAW to GND	No short
VIN to GND	No short; capacitor charging may briefly affect reading
LOGIC_3V3 to GND	No short
STANDALONE POWER	Removed while UART USB is connected
COIL POWER	Removed until the relay supply and driver are bench-tested